ScienceDirect

Ceramics International

Volume 51, Issue 23, Part B, September 2025, Pages 40212-40222

Optimization of braze sealing process for ZrO_2/Ti6Al4V glass connections: Interface impact studies

Fei Ji, Yuanxing Li, Wenxin Dong, Yuanjie Chi, Menglin Li, Lukai Luo, Tao Zou, Hui Chen

Show more

Outline | Share | Cite

https://doi.org/10.1016/j.ceramint.2025.06.254

Get rights and content

Full text access

Abstract

This study systematically investigates the interface-strengthening mechanisms of bismuth-based sealing glass for brazing Ti6Al4V titanium alloy and ZrO_2 ceramic in an air environment. As key materials in aerospace, biomedical, and electronic packaging applications, the reliable joining of titanium alloys and ceramics is significant for expanding their composite applications. Given the thermal sensitivity of sealing glass, which requires precise control of the bonding temperature to achieve reliable connections, this study explores the influence of brazing temperature and holding time on the interface structure and mechanical properties. When the brazing process parameters were optimized to 650 °C/30 min, the joint achieved the highest shear strength, with an average value of 35 MPa. Under these optimized conditions, a composite structure consisting of nanoscale $Bi_4Ti_3O_{12}$ precipitates and Bi single-phase formed at the interface, effectively enhancing the bonding strength of the heterogeneous materials through chemical bonding. When the temperature exceeded the critical value or the holding time was extended, abnormal coarsening of the precipitates occurred, leading to intensified stress concentration effects and a significant decline in interfacial bonding strength. By establishing the correlation mechanism among process parameters, interface structure, and mechanical properties, this study confirms that controlling the size of interfacial precipitates is a key factor in improving the bonding strength of metal/ceramic joints.

Keywords

ZrO_2 ceramic; Sealing glass; Brazing; Interfacial strengthening; Joint performance

1. Introduction

Since the 20th century, industrial sectors have witnessed remarkable advancements, particularly in high-tech fields, including aviation, telecommunications, and defense industries. The structural materials supporting these domains have been evolving toward achieving lighter weight, higher precision, miniaturization, and multifunctionality. As an emblem of technological progress, metallic materials have found extensive applications across numerous industries. Among these, Ti6Al4V titanium alloy stands out for its exceptional properties, combining lightweight characteristics with high strength, superior corrosion resistance, excellent tensile properties, and remarkable toughness [1,2]. Ti6Al4V alloy has been extensively utilized across various industrial sectors due to its excellent mechanical properties and corrosion resistance, demonstrating significant application potential [3,4]. However, the relatively poor wear resistance of Ti6Al4V limits its performance under certain demanding engineering conditions and specific operational environments. To address this limitation, there is a pressing need to develop complementary materials that can enhance the overall performance of Ti6Al4V and broaden its application scope. Currently, several joining techniques are available for titanium alloys, including mechanical fastening, adhesive bonding, brazing, and solid-state welding [[5], [6], [7], [8], [9]]. Among these methods, brazing has gained particular preference in industrial applications owing to its operational simplicity, process reliability, consistent product quality, and cost-effectiveness [[10], [11], [12]].

As an innovative joining material, low-temperature sealing glass enables the bonding of dissimilar materials (including metal-to-metal, metal-to-ceramic, and metal-to-glass combinations) under relatively low temperatures and atmospheric conditions. This approach not only reduces manufacturing costs but also simplifies processing procedures [13]. Sealing glass technology has found extensive applications in various high-tech industries such as automotive manufacturing, aerospace engineering, and microelectronics. It has successfully addressed numerous technical challenges across multiple advanced technology fields, including electrical engineering, electronic sensors, optical systems, structural mechanics, nuclear technology, superconductors, and microfluidic devices [[14], [15], [16]]. However, the performance of glass solder exhibits significant temperature dependence. Therefore, selecting an optimal brazing temperature is crucial to maximizing the sealing glass's effectiveness and enhancing joint performance. Wang et al. [17] investigated a Bi_2O_3-ZnO-B_2O_3 system sealing glass for joining DM305 electronic glass and Kovar alloy. Their study revealed that effective bonding was achieved through interfacial interdiffusion when the joint reached a critical temperature. At a bonding temperature of 500 °C, the joint exhibited a maximum shear strength of 12 MPa. Similarly, Lin et al. [18] utilized Bi-based sealing glass to reinforce joints via interfacial reactions at 700 °C, achieving a significantly higher shear strength of 35 MPa. Chen et al. [19] successfully achieved the bonding of nickel-plated copper and Al_2O_3 ceramics using bismuth-based sealing glass, with the joint strength peaking at 21 MPa when brazed at 680 °C. These studies collectively demonstrate that maximum joint strength is attained when the sealing glass forms distinct precipitated phases at the interface. The distribution and morphology of these precipitated phases exhibit a strong correlation with brazing parameters and play a crucial role in determining the ultimate joint performance. Consequently, to consistently optimize the mechanical strength of sealing glass joints, a comprehensive investigation into the direct relationship between brazing process parameters and joint performance is imperative.

Previous research has systematically investigated the influence of various thermal oxidation processes on the wettability between Ti6Al4V surfaces and sealing glass [20]. Building upon this foundation, the current study extends the investigation to examine how critical brazing parameters—particularly brazing temperature and holding time—affect the performance of Ti6Al4V/sealing glass/ZrO_2 joints, with a comprehensive analysis of interfacial evolution. This work aims to (1) identify the composition of strengthening phases formed at the interface, (2) elucidate their underlying strengthening mechanisms, and (3) establish theoretical guidelines for optimizing sealing glass-based joining of titanium alloys.

2. Experimental details

2.1. Materials

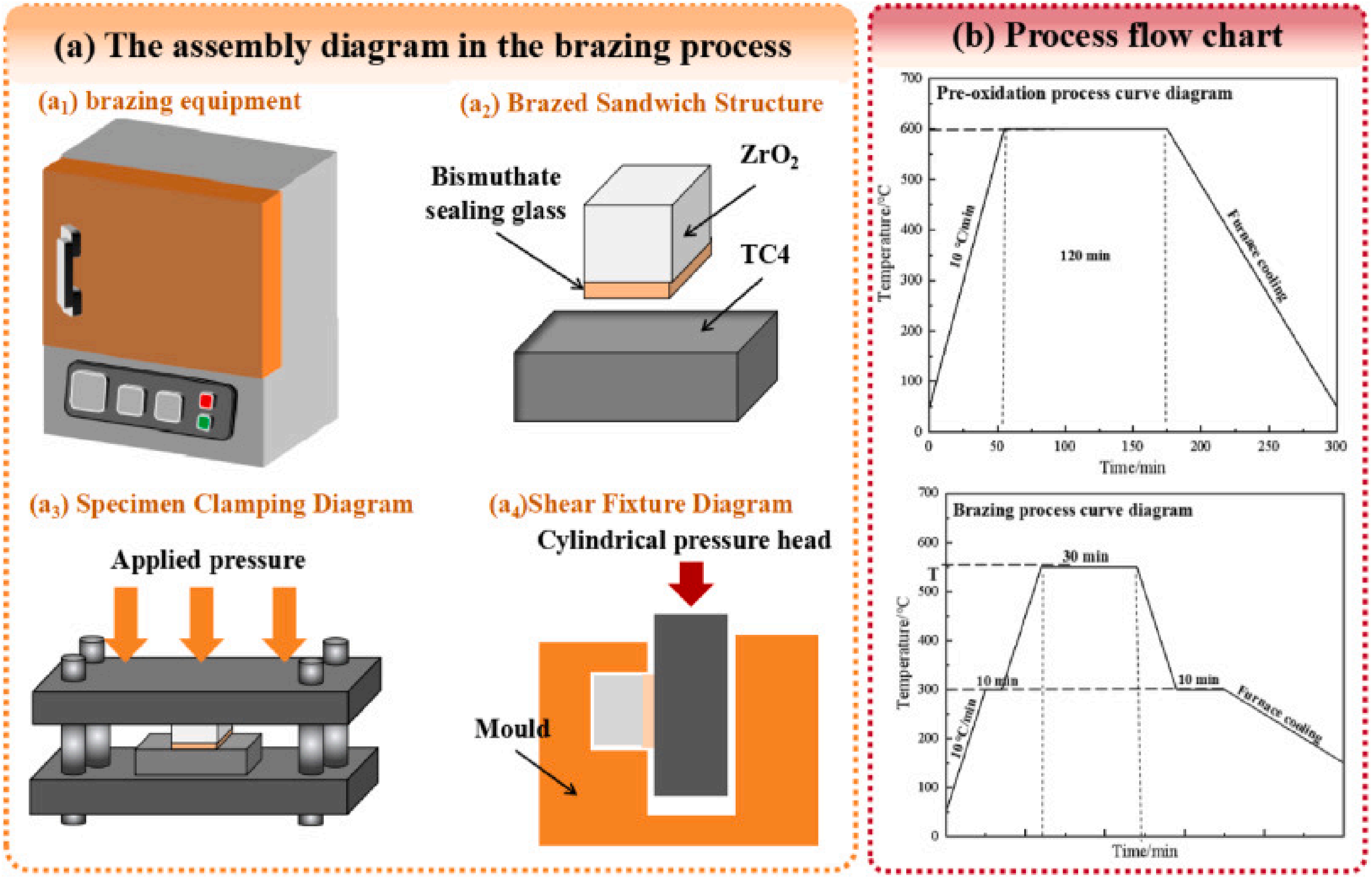
The base materials employed in this study were Ti6Al4V titanium alloy and ZrO_2 ceramic, with specimen dimensions of 20mm×10mm×5mm and 10mm×10mm×5mm, respectively. These materials were precisely sectioned using a laser cutting machine. Before surface treatment, the specimens were immersed in an anhydrous ethanol solution and subjected to degreasing treatment through ultrasonic cleaning in an ultrasonic oscillation cleaner. The sealing glass used was a bismuthate-based sealing glass supplied by Shijiazhuang Low-melt Glass Technology Co., Ltd. The preparation process of the sealing glass involved high-temperature melting followed by water quenching treatment. The resulting glass blocks were ground into powders with an average particle size below 5μm. The chemical composition and content of the sealing glass were precisely characterized using X-ray fluorescence (XRF), with detailed parameters presented in Table 1. Furthermore, the sealing glass was confirmed to be amorphous with excellent structural stability, exhibiting a glass transition temperature (Tg) of 443°C and a softening temperature (Tf) of 482°C.

Table 1. Composition of the bismuthate sealing glass (wt%).

Compound	**Bi_2O_3**	**ZnO**	**SiO_2**	**Na_2O**	**Al_2O_3**	**BaO**	**TiO_2**	**ZrO_2**
Concentration	75.03%	6.89%	6.54%	3.43%	3.02%	2.18%	1.79%	1.12%

2.2. Brazing method

In this study, a high-temperature box-type resistance furnace was employed to conduct pre-oxidation treatment and joining experiments on titanium alloy substrates. The specimen assembly configuration is illustrated in Fig. 1(a), which depicts the layout of Ti6Al4V and ZrO_2 during the welding process, along with the fixture model used for testing the shear strength of the joints.



Download: Download high-res image (935KB)

Download: Download full-size image

Fig. 1. Brazing processing schematic diagram (a) and pre-oxidation and brazing processing curve diagram (b).

The sealing glass was mixed with terpineol at 3:1 to prepare the brazing paste. The paste was uniformly applied to the material surfaces using a screen printing technique and allowed to dry at room temperature. The oxidation process and brazing procedure are detailed in Fig. 1(b), with the thermal cycle encompassing the heating, holding, and cooling stages. The brazing temperature T was determined based on the softening temperature of the sealing glass, while the heating rate was set to 10°C/min. Based on previous studies referenced in this research, it was determined that the pre-oxidation treatment at **600°C/120 min** yielded optimal wettability of the sealing glass. Consequently, this oxidation process was adopted in the current study to oxidize the alloy side. Combined with the glass properties, the viscosity of the sealing glass during full spreading must reach **10^3–10^5 Pas**, and the joining temperature typically needs to be **approximately 100°C higher than the softening temperature** (482°C). Therefore, the brazing process parameters were selected by considering these characteristic requirements, with specific conditions detailed in Table 2.

Table 2. Brazing connection process table.

NO.	brazing temperature (°C)	holding time (min)	Glass Solder
1	550°C	30	paste
2	600°C		
3	650°C		
4	700°C		
5	650°C	60	
6		90	

Table 3. EDS point scanning data at typical locations in Fig. 4(d).

point	Ti	Al	V	Bi	Si	Zn	Na	Zr	O	produced phase (prediction)
A	13.66	1.08	0	12.44	1.44	2.39	2.18	0	66.81	$Bi_4Ti_3O_{12}$
B	16.08	0.75	0.27	12.51	1.04	2.15	2.07	0	65.13	$Bi_4Ti_3O_{12}$
C	27.75	1.89	0.59	1.04	0.64	0.26	1.34	0	66.49	TiO_2
D	33.01	0.37	0.33	0.73	0.02	0	0.13	0	65.41	TiO_2

point	Ti	Al	V	Bi	Si	Zn	Na	Zr	O	produced phase (prediction)
E	1.49	0.22	0.52	70.64	0.18	0.33	0	0	26.62	Bi single phase

2.3. Characterization

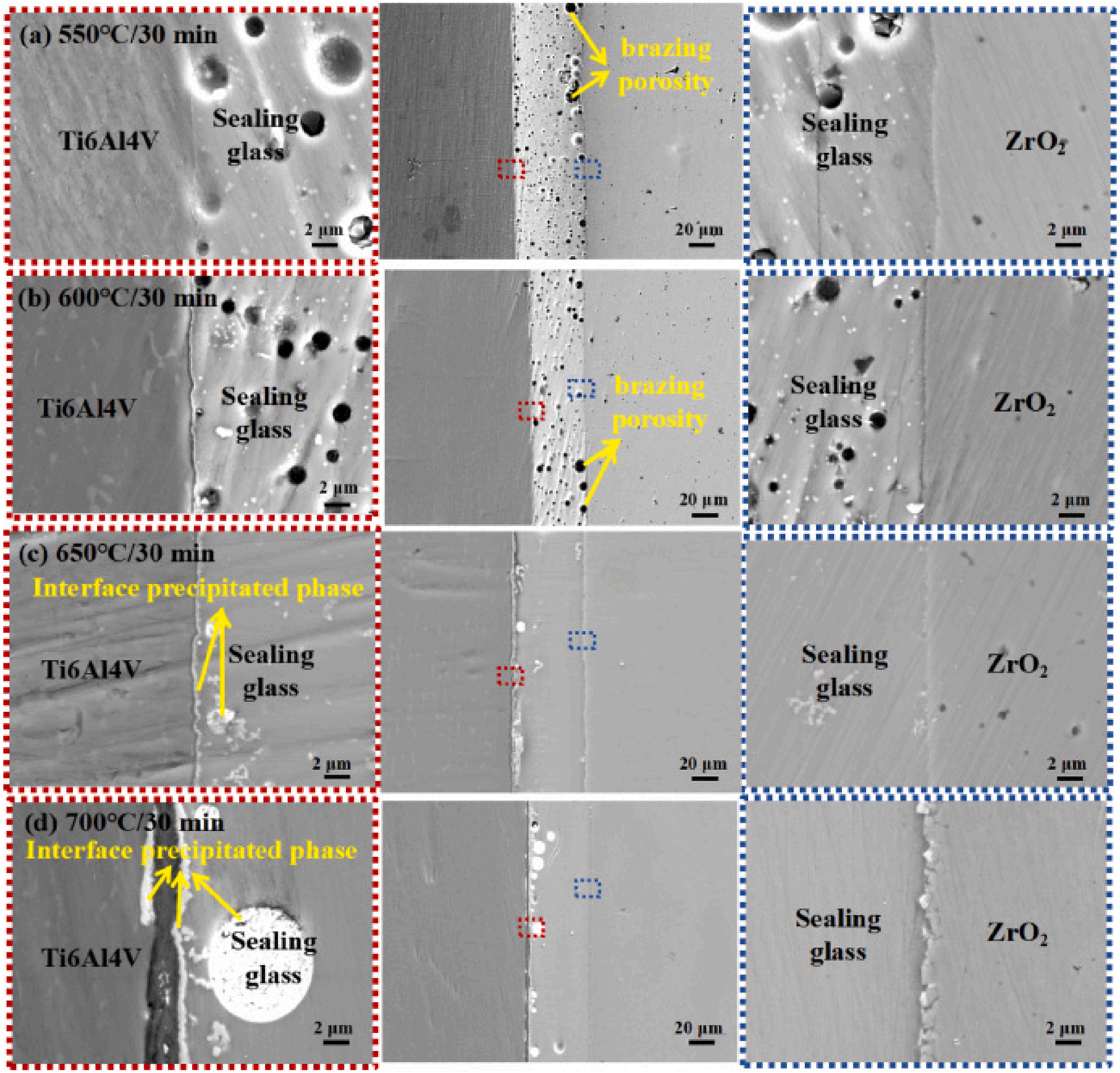
The microstructural morphology of Ti6Al4V/ZrO_2 joints was characterized using a scanning electron microscope (SEM, GeminiSEM 300, Zeiss, Germany) (see Table 3). Elemental diffusion across the joint interface was analyzed via surface mapping in energy dispersive spectroscopy (EDS, SmartEDX, Germany). A field-emission transmission electron microscope (TEM, FEI Talos F2000X) with an accelerating voltage of 200 kV and a resolution ≤0.12 nm was employed to investigate characteristic diffusion layers in the joints. Bright-field imaging was utilized to observe the elemental distribution within diffusion layers, while high-resolution TEM (HRTEM) and selected area electron diffraction (SAED) were applied to further analyze distinct precipitated phases. TEM specimens were prepared using the lift-out mode of a focused ion beam (FIB, FEI Helios NanoLab 600i).

3. Results

3.1. Surface microstructure and element distribution

Given that sealing glass, as a bonding material with thermal softening properties, exhibits a high dependence of its bonding performance on temperature variations during the brazing process. As the temperature fluctuates, the viscosity of sealing glass undergoes corresponding changes. When the brazing temperature gradually increases, the sealing glass first reaches its transition temperature. At this specific temperature threshold, significant transformations occur in the physical and chemical structures within the glass, enabling it to achieve a flow state. With further temperature elevation, the glass enters its softening temperature range. Within this range, the glass transitions from a hardened to a softened state, accompanied by notable structural changes characterized by an increase in density and a significant reduction in volume, ultimately achieving a fused state.

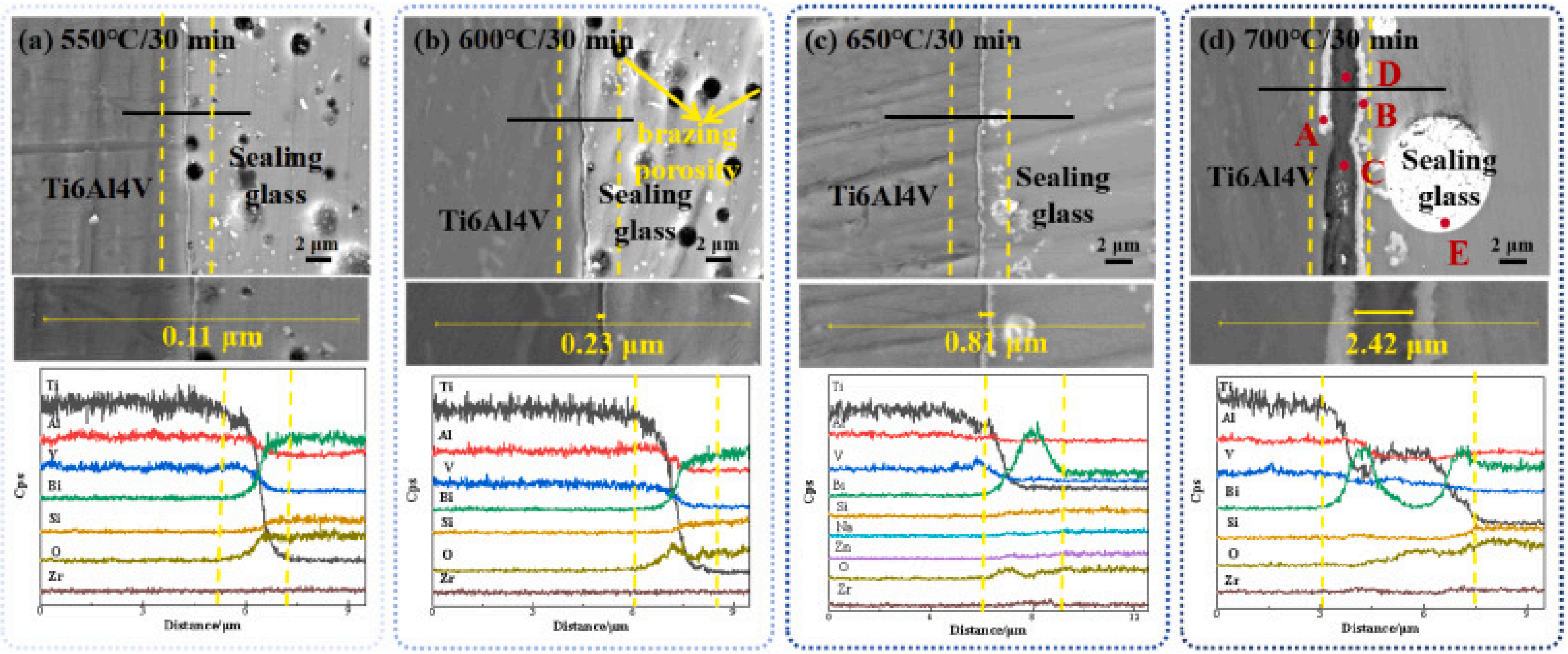
The joint morphologies of Ti6Al4V/sealing glass/ZrO_2 at different sealing temperatures are shown in Fig. 2, illustrating variations at both interfaces and within the sealing glass under various brazing temperatures. Through detailed observation of the microstructure at the bonding interface and adjacent regions, it was observed that at lower temperatures (550 °C and 600 °C), numerous pores existed within the sealing glass. This phenomenon is attributed to the fact that the sealing glass was prepared from an organic solvent-containing paste-like brazing filler. Within this temperature range, the organic solvent failed to volatilize fully, resulting in residual pores of varying sizes in the weld seam. Notably, the pore sizes at 600°C were generally smaller than those at 550°C. As the welding temperature increased to 650°C and 700°C, the brazing filler within the weld seam exhibited a uniform, pore-free state accompanied by white circular particulate precipitates. Further analysis of the interfacial microstructure between the titanium alloy side and the sealing glass at different brazing temperatures revealed that as the temperature increased, the dark interfacial layer gradually became more distinct and significantly thickened. Concurrently, pronounced white precipitates emerged on the sealing glass side, with their thickness increasing with temperature. When the temperature reached 700°C, the white precipitated phase diffused across the dark interfacial layer into the titanium alloy substrate side (see Fig. 3).



Download: Download high-res image (2MB)

Download: Download full-size image

Fig. 2. Joint morphology at different brazing temperatures (a) 550°C; (b) 600°C; (c) 650°C; (d) 700°C.



Download: Download high-res image (1MB)

Download: Download full-size image

Fig. 3. The interface microstructure and element distribution of Ti6Al4V/sealing glass at different brazing temperatures were studied. (a) 550°C; (b) 600°C; (c) 650°C; (d) 700°C.

Through meticulous observation and analysis of the effects of temperature variations on the microstructure of joints, it was found that the sealing glass began to exhibit uniform filling effects at a temperature setting of 650°C, while the precipitated white spherical phases also demonstrated improved uniformity in both size and distribution. Based on this critical discovery, this brazing temperature was selected as the benchmark to conduct an in-depth comparative study on the morphological evolution of joints under different holding times. Fig. 4 illustrates the microstructural changes at the joint and interface under varying holding times through detailed characterization of microscopic features.

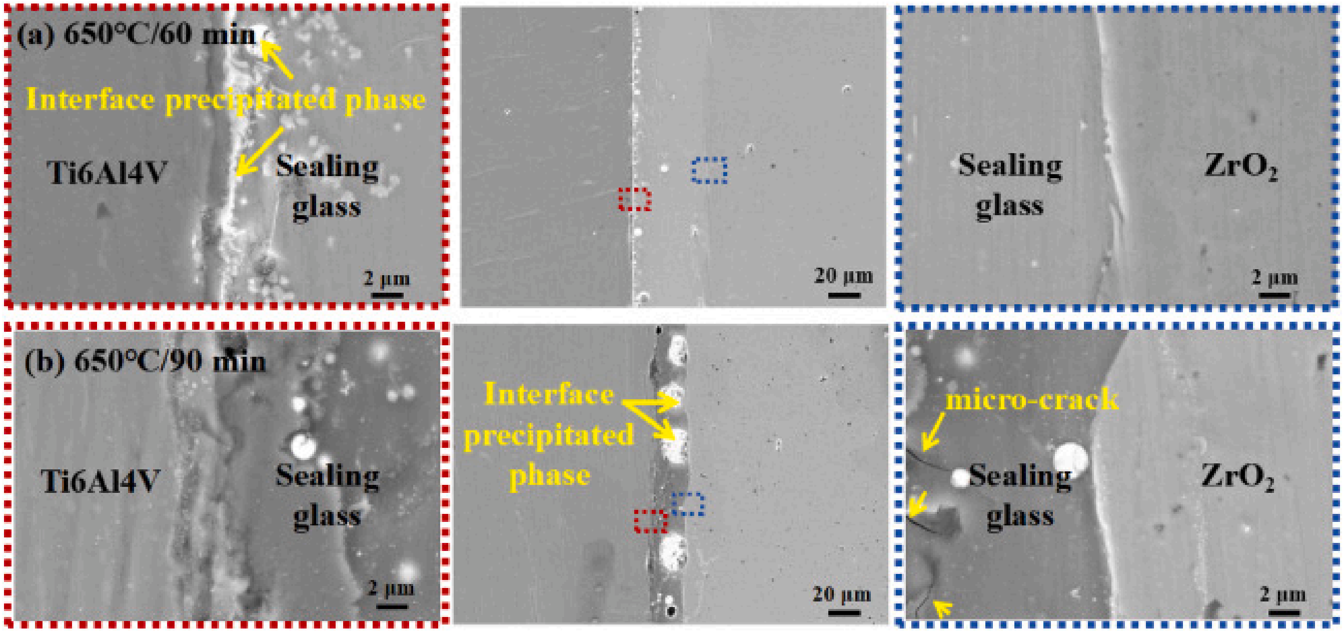
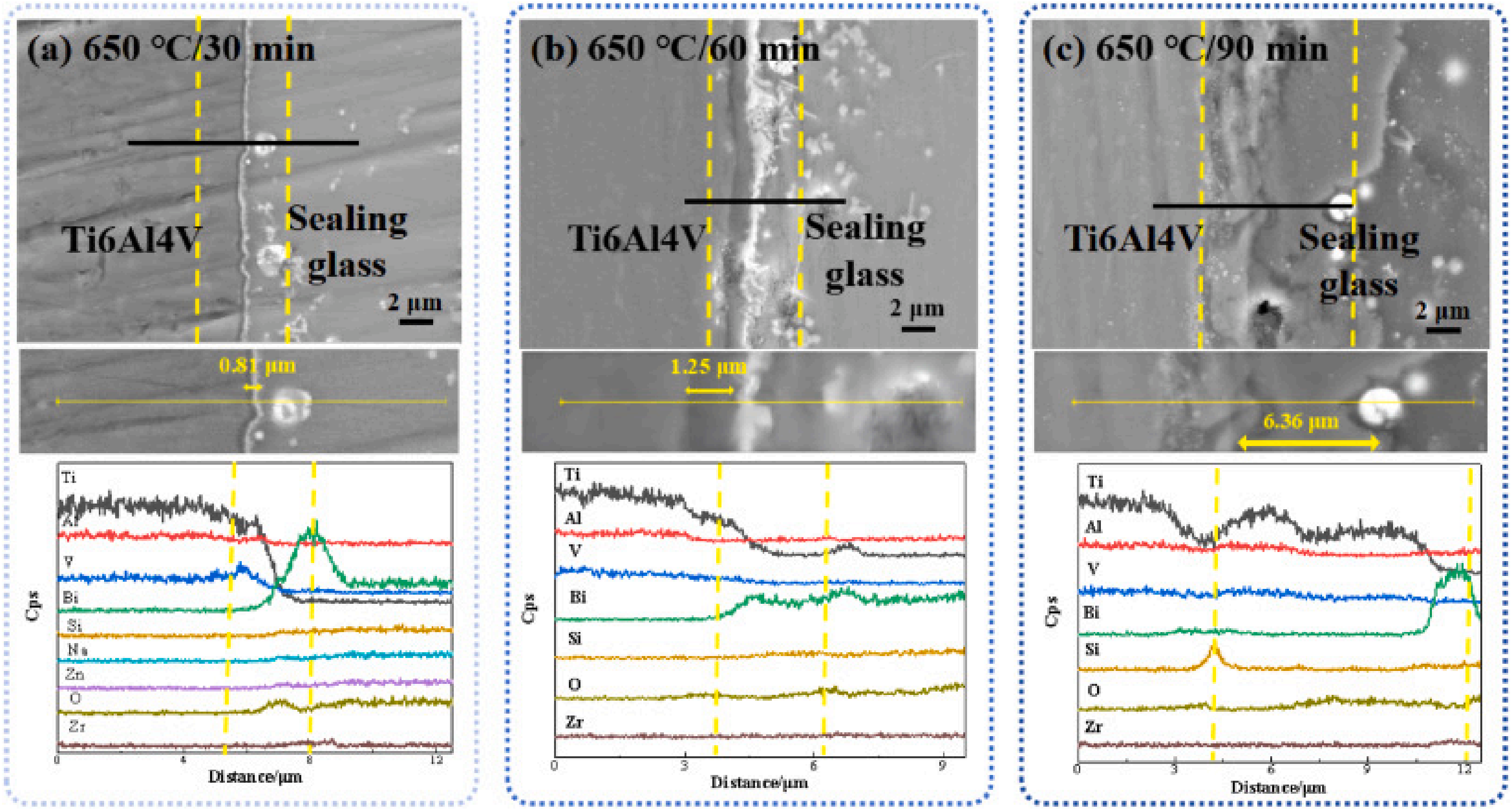


Fig. 4. The joint morphology under different holding time (a) 60 min and (b) 90 min.

SEM observations revealed that when the holding time was extended to 60 min, the sealing glass layer became homogeneous and pore-free, the oxide layer on the titanium alloy side significantly thickened, and the diffusion layer between the oxide film and sealing glass further expanded, with the density of precipitated phases correspondingly increasing. However, when the holding time exceeded 90 min, the color of the brazing layer darkened noticeably, and the precipitated phases along the alloy/brazing layer interface disappeared, replaced by white ellipsoidal precipitates formed by Bi element aggregation. Additionally, cracks were observed in the weld seam. The comparative analysis further indicated that the Bi-based spherical precipitates near the titanium alloy/sealing glass interface gradually increased in size and shifted toward the ceramic side with prolonged holding time. In this study, the EDS line scanning method was used to analyze the distribution of interface elements after different holding time treatments. Fig. 5 shows the corresponding data and reveals the variation of element distribution at the interface with the holding time.



Download: Download high-res image (1MB)

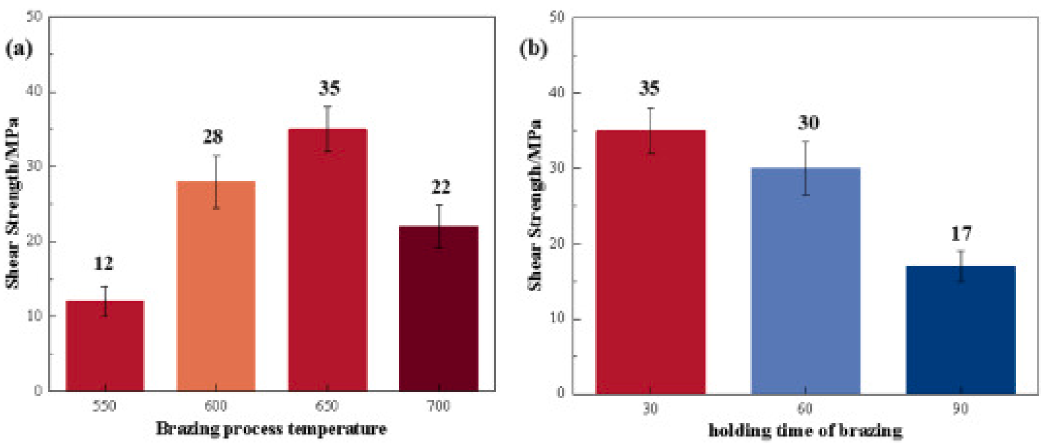
Download: Download full-size image

Fig. 5. The interface microstructure and element distribution of Ti6Al4V/sealing glass under different heat preservation conditions were studied. (a) 30 min; (b) 60 min; (c) 90 min.

Through meticulous EDS mapping analysis, it was observed that the content of Bi-dominated precipitated phases in the interfacial region significantly increased with prolonged holding time. Additionally, numerous white dot-like precipitated phases and high-purity Bi elemental microspheres emerged within the sealing glass. Specifically, when the holding time was set at 30 min and 60 min, the oxide film layer on the titanium alloy surface exhibited uniformity and demonstrated remarkable interdiffusion with the sealing glass. This interdiffusion resulted in the formation of characteristic precipitated phases at the interface. However, as the holding time extended to 90 min, the oxide film thickness further increased while displaying nonuniform characteristics. Notably, under this prolonged holding condition, no distinct precipitated phases were detected at the interface. The bonding quality at the interface serves as a critical determinant of joint performance. Based on a comprehensive analysis of interfacial morphology and elemental distribution, it can be concluded that excessive holding time exerts detrimental effects on the bonding strength between the sealing glass and titanium alloy oxide film, thereby adversely impacting the overall mechanical properties of the joint. Consequently, optimizing the holding duration proves essential for refining sealing processes and enhancing joint quality.

3.2. Joint performance and fracture morphology

Sealing glass composed of oxides exhibits significant brittle characteristics as its internal atoms are bonded through covalent and ionic interactions. Shear strength serves as a crucial parameter for evaluating joint performance. The fracture morphology is collectively influenced by multiple factors, including the composition of the brazing filler material, microstructural characteristics, and testing conditions. Fig. 6 reveals the shear strength values under different brazing process parameters. The research results demonstrate that the maximum shear strength, with an average value of 35 MPa, is achieved under the brazing condition of 650 °C/30 min. Through microstructural analysis of the joint, a correlation has been identified between the size and morphology of spherical precipitates in the weld seam and joint performance. Comparative analysis indicates that when the size of precipitates exceeds a specific range - whether excessively large or too small - it adversely affects the strength performance of the joint.

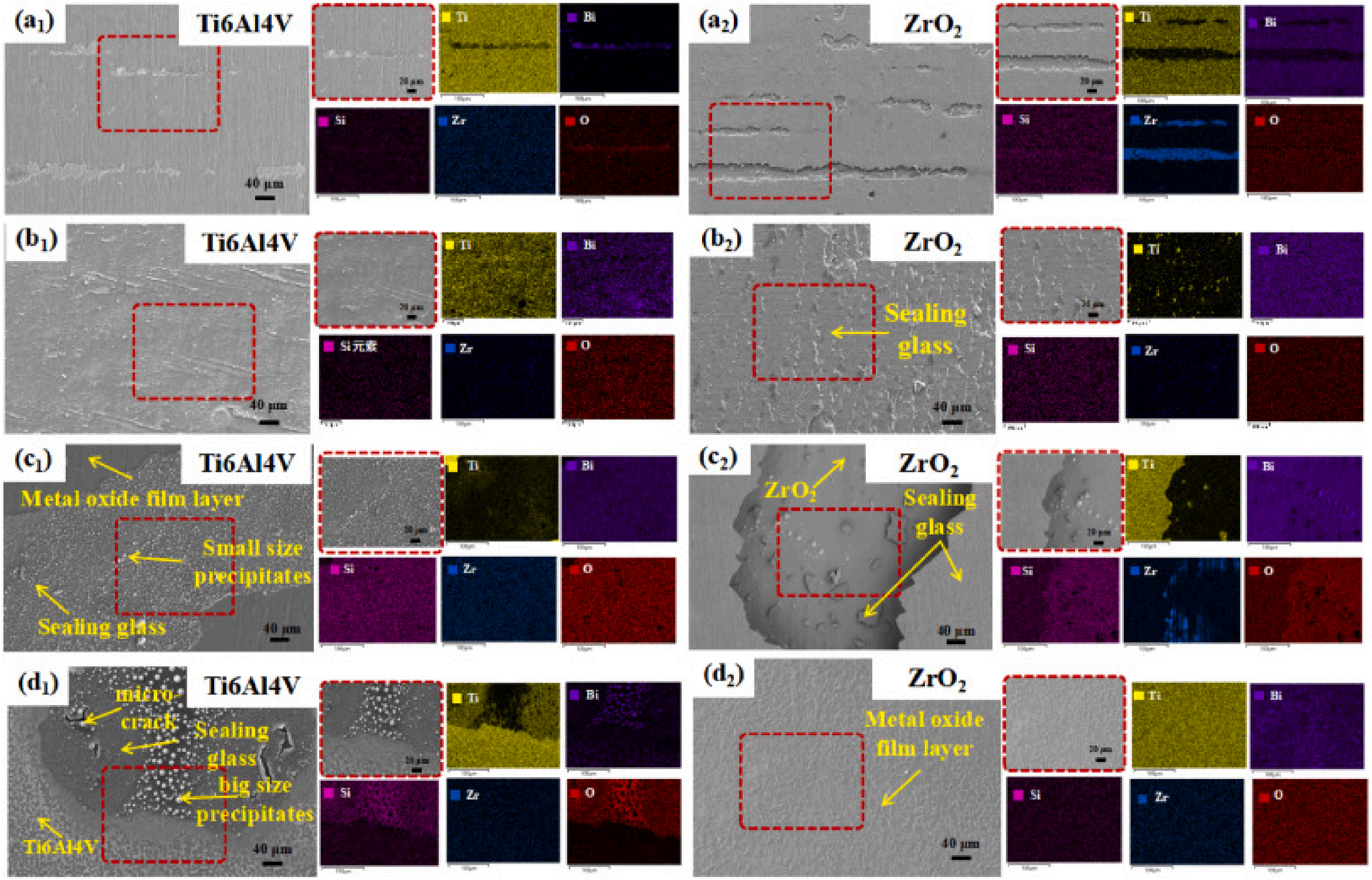


Download: Download high-res image (208KB)

Download: Download full-size image

Fig. 6. Shear resistance of joints (a) different brazing temperature and (b) different holding time.

Fig. 7 presents the EDS-Mapping spectra of fracture morphologies and locations under different brazing temperatures. The study revealed that at a brazing temperature of 550°C, a large area of brazing filler metal adhered to the ceramic side, while only a tiny amount of filler metal remained attached to the metal side. The fracture surface exhibited smooth characteristics, indicating poor bonding between the metal side and the sealing glass at this temperature. When the temperature increased to 600°C, layered fracture textures were observed at the fracture interface, attributed to brittle fracture of the glass. Although most filler metal still adhered to the ceramic side, significant changes in fracture morphology occurred, suggesting an improvement in fracture strength. Upon further increasing the brazing temperature to 650°C, fractures primarily occurred within the filler metal, with filler metal adhering to both base materials and accompanied by the appearance of white spherical precipitates. At 700°C, numerous larger white spherical particles formed within the filler metal on the titanium alloy side, while an alloy oxide layer was observed on the ceramic fracture surface. The analysis suggests that the excessive thickness of the oxide film on the metal surface caused by high-temperature brazing weakened the film-substrate bonding strength, resulting in fractures occurring between the alloy substrate and the oxide film.

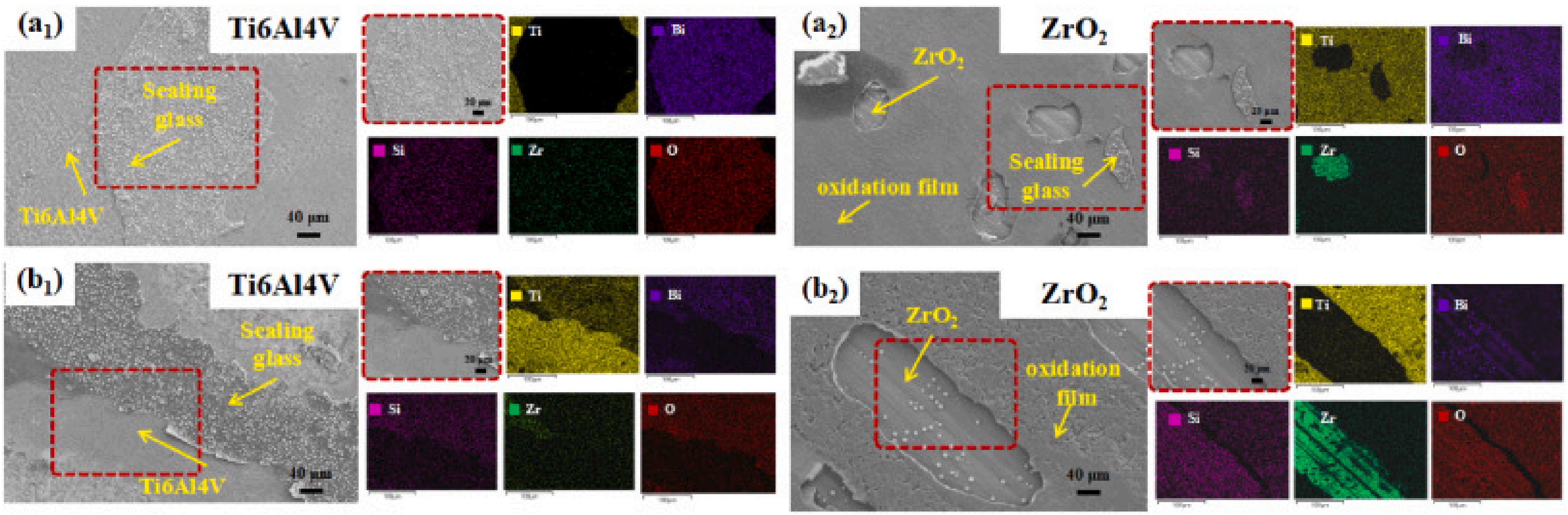


Download: Download high-res image (3MB)

Download: Download full-size image

Fig. 7. The fracture morphology of the joint at different brazing temperatures (a) 550°C; (b) 600°C; (c) 650°C; (d) 700°C.

Based on the aforementioned analysis, while brazing temperature significantly influences the performance of welded joints, holding time also serves as a critical factor in the brazing process. This study, therefore further comparatively analyzed the effects of holding time on joint performance, with the fracture morphology and EDS-Mapping spectra shown in Fig. 8. Experimental observations revealed that when the holding time was extended to 60 min, the ceramic side exhibited a substantial increase in filler metal adhesion along with relatively smooth fracture surfaces. However, a significant number of spherical precipitates remained observable at the metallic-side fracture interface, suggesting that the formation of these precipitates may contribute to strengthening the joint performance. When the holding time was further prolonged to 90 min, distinct filler metal edge detachment and pronounced internal cracks were observed in the metallic-side fracture surfaces, whereas the ceramic side displayed extensive adherent deposits. EDS mapping analysis confirmed that fracture locations predominantly occurred at the interface between the oxide film layer and the alloy substrate. This fracture characteristic aligns with the failure mode observed when the welding temperature reached 700 °C, indicating analogous interfacial weakening mechanisms under prolonged thermal exposure.



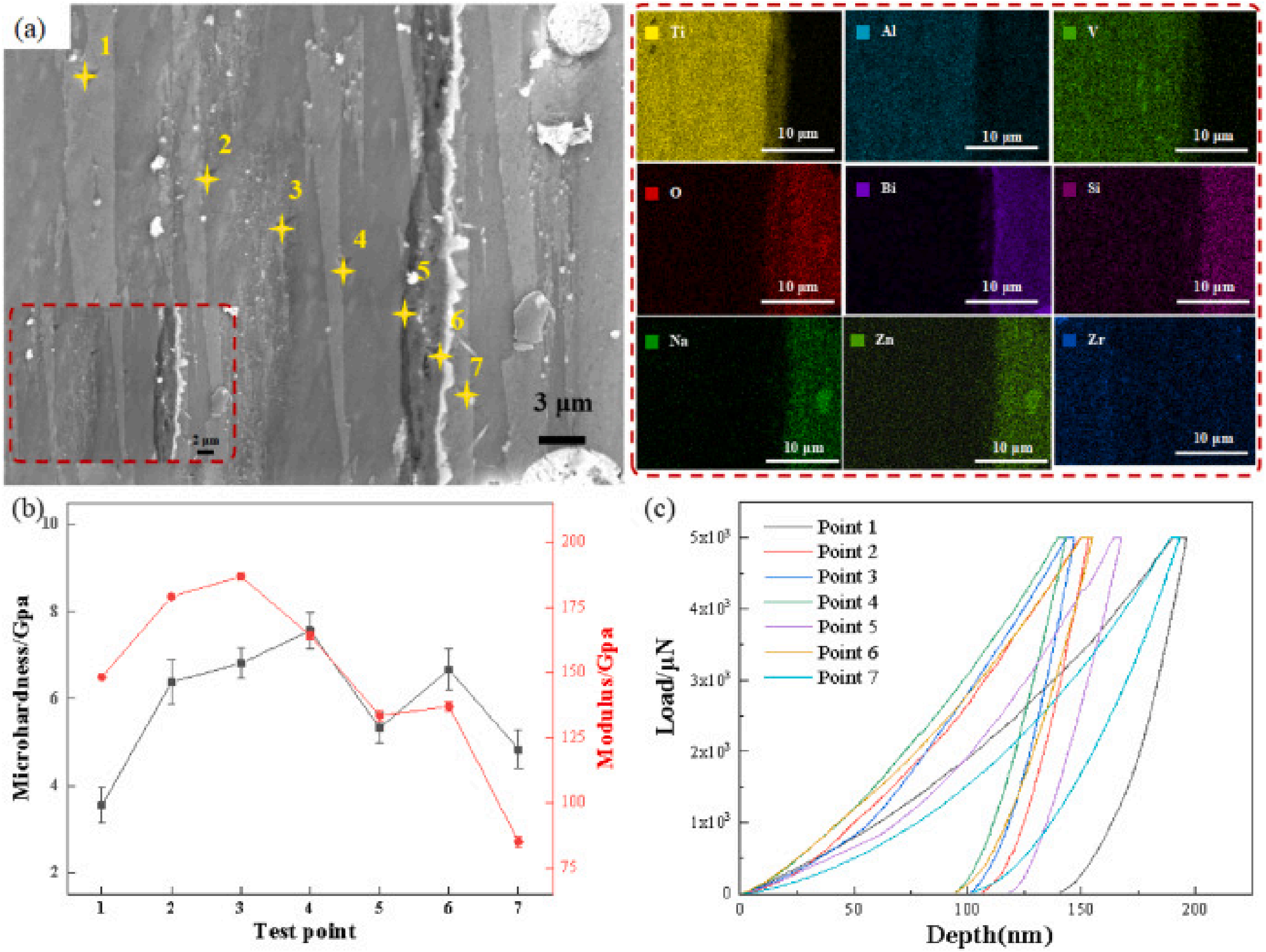
Download: Download high-res image (1MB)

Download: Download full-size image

Fig. 8. Fracture morphology and EDS-Mapping map of joint under different holding time (a)60 min and (b) 90 min.

3.3. Study on interface mechanism

To analyze the bonding mechanism at the interface further, an in-depth investigation was conducted on interfaces under characteristic parameters. Since the thickness of precipitated phases and the size of spherical precipitated phases reached their maximum when the brazing holding time was 30 min and the bonding temperature achieved 700 °C, the interface under these process parameters was selected for thorough analysis. This study systematically characterized the mechanical properties of the surface-modified layer on Ti6Al4V alloy using nanoindentation technology. Fig. 9 presents the interfacial morphology between Ti6Al4V and sealing glass under the brazing process, along with corresponding nanoindentation data.



Download: Download high-res image (2MB)

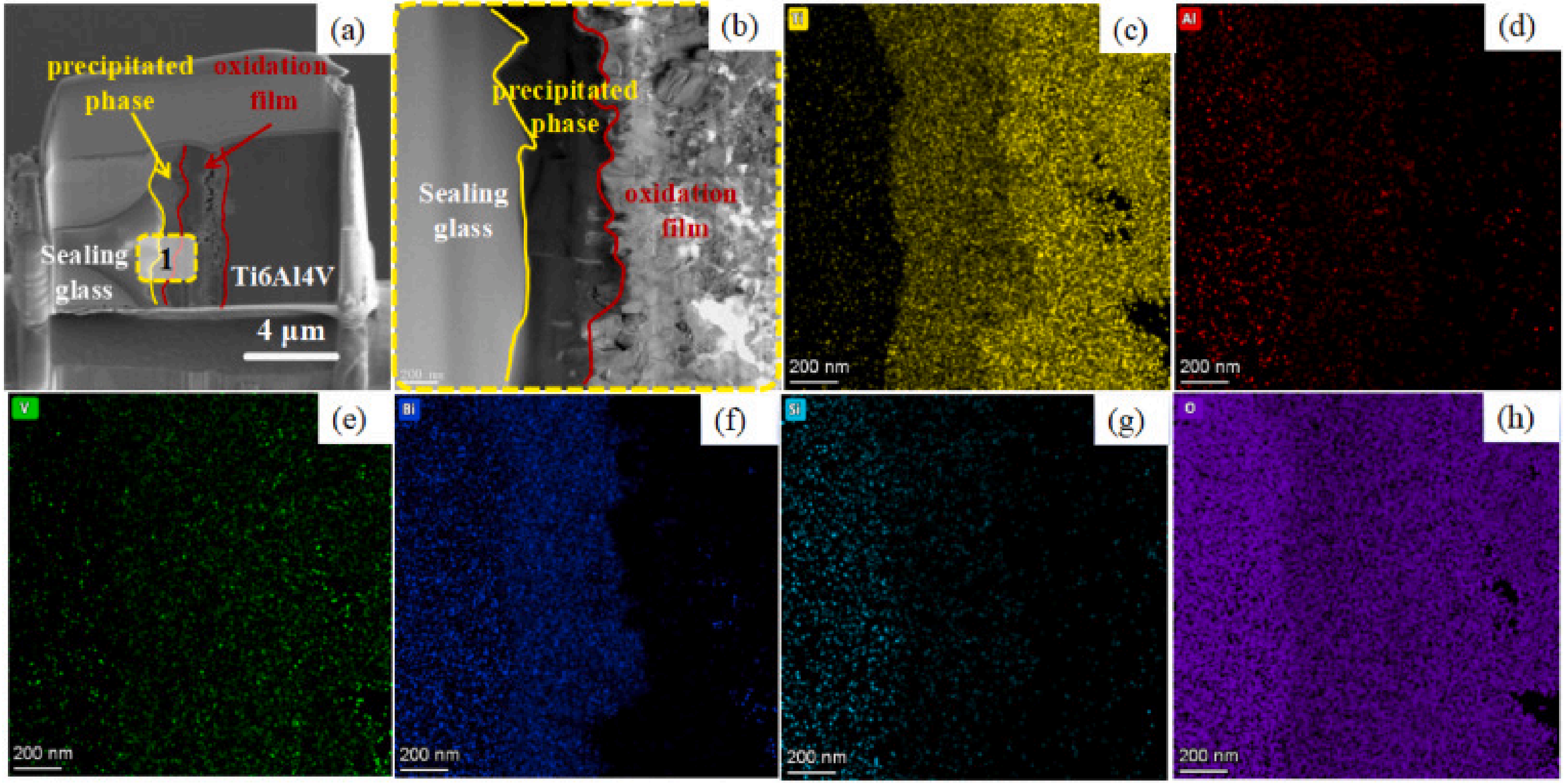
Download: Download full-size image

Fig. 9. The nanoindentation results of Ti6Al4V/sealing glass interface under the brazing process of 700°C/30min (a) Micro-morphology and element distribution; (b) Nano-hardness and Young's modulus of the test point; (c) Select the load-depth curve

of the test.

The test regions were categorized into three characteristic zones based on material composition differences: the original alloy matrix zone (points 1–3), the thermally oxidized film transition zone (points 4–5), and the sealing glass layer (point 6). A Berkovich diamond indenter was employed for testing, with loading parameters set at 5 mN load and 2 s holding time to ensure reliable nano-hardness data. Test results of the matrix zone revealed that the Ti6Al4V matrix exhibited a mean nano-hardness of 6.95 ± 0.32 GPa (n = 3). Notably, a peak hardness value of 7.60 GPa was observed in the parent metal region, representing a 9.4 % increase compared to the matrix average. The oxidized transition layer demonstrated moderate mechanical properties, with a mean nano-hardness of 6.26 ± 0.55 GPa (n = 2), showing a 9.9 % reduction relative to the matrix but remaining significantly higher than the sealing glass layer's 4.93 GPa (21.1 % decrease). The gradient structural design of the material system exhibited remarkable mechanical coordination advantages: A continuous hardness gradient was formed from the matrix (6.95 GPa) → oxidized layer (6.26 GPa) → glass layer (4.93 GPa), accompanied by a synchronous stepwise decline in elastic modulus (matrix ~177 GPa → oxidized layer ~136 GPa → glass layer ~84 GPa). This gradual mechanical performance distribution effectively alleviates stress concentration at heterogeneous material interfaces.

FIB cutting was performed on the interface of nanoindentation analysis in the above figure. Fig.10 (a) is the FIB interface section, and region 1 in the figure was selected for TEM EDS-Mapping analysis, as shown in Fig. 10 (c) ~ (h). It is found that the oxide film layer of Ti6Al4V and sealing glass directly diffuses with each other, forming a new interface layer, which is composed of Ti, Bi and O elements.

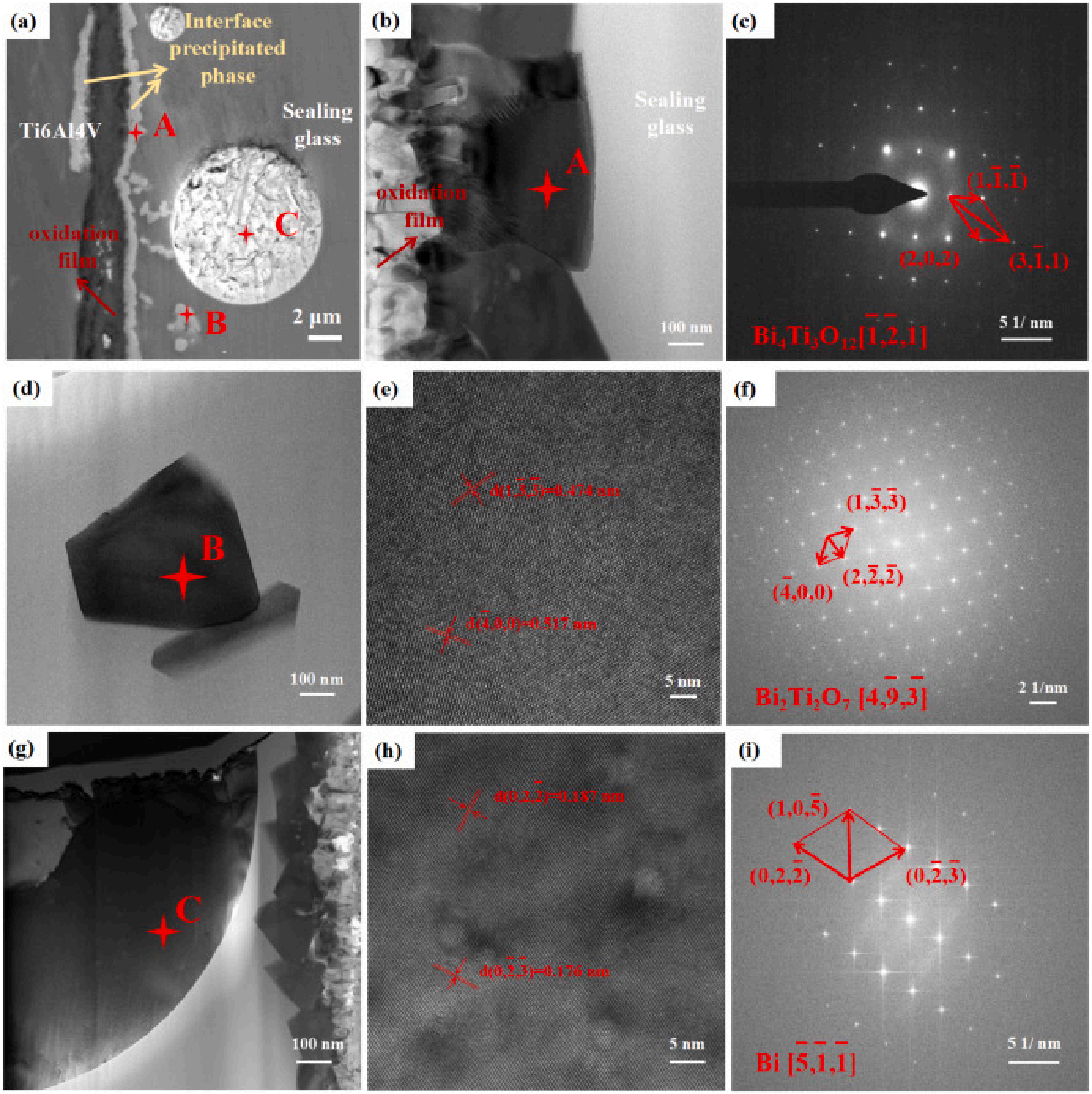


Download: Download high-res image (2MB)

Download: Download full-size image

Fig. 10. 700°C/30min brazing process of Ti6Al4V/sealing glass interface HAABF-TEM microscopic image (a) FIB cross-section; (b) The bright field diagram of region b; (c)–(h) Mapping map.

This study systematically characterized the precipitated phases at the Ti6Al4V/sealing glass interface after 700°C/30min heat treatment using transmission electron microscopy (TEM). As shown in Fig. 11(a), three typical morphologies of precipitated phases were observed in the interfacial region: acicular phase (A), blocky phase (B), and spherical phase (C). Multi-dimensional microstructural analyses revealed the crystal structures and formation mechanisms of these distinct phases.



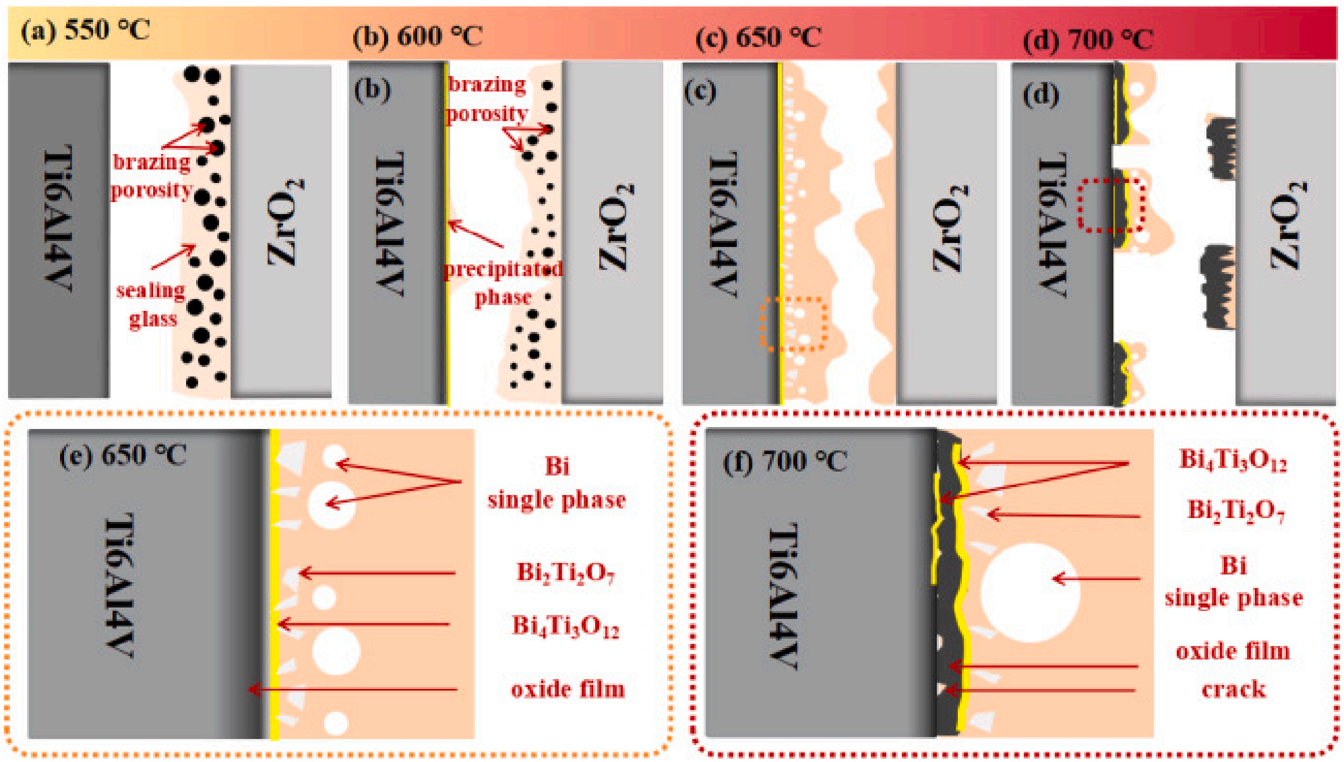
Download: Download high-res image (3MB)

Download: Download full-size image

Fig. 11. TEM of Ti6Al4V/sealing glass interface layer at 700°C/30min: (a) Interface microstructure; (b) The bright field image of region A; (c) The electron diffraction pattern of region A; (d) The bright field image of region B; (e) Amplified dark field image in region B; (f) Electron pattern diffraction pattern of B region; (g) Bright field image of region C; (h) C-region amplified dark field image; (i) Electron pattern diffraction pattern in region C.

For the rod-shaped phase (A) at the interface boundary, high-angle annular dark-field TEM (HAADF-TEM) combined with selected-area electron diffraction (SAED) was employed for detailed investigation. Fig. 11(b) demonstrates its characteristic layered growth pattern, with corresponding diffraction patterns (Fig. 11(c)) indexed as $Bi_4Ti_3O_{12}$. This compound crystallizes in an orthorhombic layered perovskite structure (space group B2cb), featuring alternating stacking of $[Bi_2O_2]^{2+}$ layers and perovskite-type $[Ti_3O_{10}]^{2-}$ layers [[21], [22], [23], [24]]. This unique layered structure contributes to interfacial stress alleviation and bonding strength enhancement. In-depth analysis of the blocky phase (B) revealed its close correlation with reaction kinetics. As illustrated in Fig. 11(d)–(f), high-resolution TEM (HRTEM) and fast Fourier transform (FFT) analyses identified this phase as cubic pyrochlore-structured $Bi_2Ti_2O_7$ (space group Fd $\bar{3}$m). Notably, this metastable phase formation directly correlates with insufficient Bi diffusion in the system, existing as an intermediate phase during solid-state reactions. Its emergence provides critical insights into heterogeneous interface reaction kinetics. Fig. 11(g)–(i) present the refined structural characteristics of the grain-boundary spherical phase (C). HRTEM imaging revealed lattice spacings of 0.187 nm, perfectly matching the (0,2,-2) planes of metallic Bi. Combined with zero-loss filtered electron diffraction pattern indexing, this spherical phase was confirmed as a single-crystalline Bi phase [25,26].

Based on the analysis of weld microstructure, joint properties, and interface layers through transmission electron microscopy (TEM), this study elucidates the interfacial strengthening mechanism and illustrates the specific strengthening process in Fig. 12. The research reveals that insufficient welding temperature and holding time (as shown in Fig. 12(a) and (b)) result in incomplete volatilization of organic solvents in the paste filler metal, leading to numerous pores in the weld seam. These defects predominantly formed at the metal/glass sealing interface, where fractures typically occurred. With increased welding temperature, pore dimensions decreased, and uniformly distributed $Bi_4Ti_3O_{12}$ precipitates emerged at the alloy/glass sealing interface. When the welding temperature reached a critical threshold (Fig. 12(e)), complete pore elimination was achieved with enhanced filler metal homogeneity. Spherical single-crystalline Bi particles were observed near the alloy/glass interface, accompanied by significantly thickened $Bi_4Ti_3O_{12}$ phases and rhombic $Bi_2Ti_2O_7$ precipitates (identified as metastable phases) at interfacial edges. The fracture path shifted to within the sealing glass layer (Fig. 12(c)), demonstrating superior interfacial bonding on both sides. However, excessive temperature or prolonged holding time induced enlarged voids in the oxide layer, enabling precipitate diffusion into the alloy substrate. Concurrently, increased oxide layer thickness and weakened oxide/substrate interfacial bonding redirected fracture paths to the metal/oxide interface, substantially degrading joint strength. This investigation concludes that selecting optimal joining parameters is crucial for enhancing joint performance through interface optimization.



Download: Download high-res image (866KB)

Download: Download full-size image

Fig. 12. Interface bonding mechanism diagram at different bonding temperatures (a) 550°C; (b) 600°C; (b) 650°C; (d) 700°C; (e) Ti6AL4V/glass interface at 650°C; (f) Ti6AL4V/glass interface at 700°C.

4. Conclusions

In summary, this study systematically discussed the effect of the brazing process on the bonding interface of the ZrO_2/Ti6Al4V sealing glass connection. Through the analysis of the morphology, properties, and interface bonding mechanism of the joint, the following conclusions were drawn.

(1) The sealing process significantly influences the joint morphology, with critical factors including brazing temperature and holding time. Research findings reveal that at lower temperatures (550°C and 600°C), the sealed glass contains a significant number of pores due to incomplete volatilization of organic solvents. As the temperature increases to 650°C and 700°C, the filler material fills uniformly with complete pore elimination, forming uniformly distributed white spherical precipitated phases. When maintained at 60 min holding time, the interfacial oxide layer thickens with a homogeneous distribution of precipitated phases. However, extending the holding time to 90 min induces interfacial cracks, abnormal growth, and deviation of precipitated phases, ultimately degrading bonding quality.

(2) The sealing process parameters critically affect joint performance. Under the 650°C/30 min condition, the joint achieves the highest shear strength (35 MPa), attributed to the uniform distribution of interfacial precipitated phases and effective stress buffering. However, excessive temperature (700°C) or prolonged holding time (90 min) leads to the thickening of the oxide layer and reduced shear strength. At 550°C, smooth fracture surfaces indicate poor bonding integrity. In contrast, fractures occurring within the filler material at 650°C demonstrate the strengthening effect of precipitated phases. At 700°C with extended holding time, fractures propagate along the oxide layer/substrate interface, revealing interfacial weakening. This systematic evolution in fracture behavior directly correlates with microstructural changes induced by thermal conditions.

(3) EDS and TEM analyses revealed mutual diffusion of Ti and Bi elements at the interface, forming reinforcing phases such as $Bi_4Ti_3O_{12}$ and $Bi_2Ti_2O_7$, which contribute to stress alleviation and bonding strength enhancement. Nanoindentation testing demonstrated a gradient distribution of hardness and elastic modulus in the interfacial region, effectively reducing interfacial stress. The study identified that excessively high temperatures or prolonged holding times induce abnormal growth of precipitated phases and excessive thickening of the oxide layer, thereby compromising interfacial bonding integrity. Consequently, optimizing the brazing temperature and holding time to 650°C and 30–60 min is critical for improving joint performance.

(4) This study provides a process optimization basis for the high-reliability joining of ZrO_2/Ti6Al4V heterogeneous materials and reveals the correlation mechanism between interfacial reaction kinetics and mechanical properties, which has important reference value for precision packaging technology in the field of aerospace and energy equipment.

CRediT authorship contribution statement

Fei Ji: Writing – review & editing, Writing – original draft, Data curation. **Yuanxing Li:** Supervision, Resources, Funding acquisition. **Wenxin Dong:** Data curation. **Yuanjie Chi:** Data curation. **Menglin Li:** Investigation, Data curation. **Lukai Luo:** Data curation. **Tao Zou:** Data curation. **Hui Chen:** Supervision, Funding acquisition.

Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

Acknowledgments

This study is supported by the National Natural Science Foundation of China (Nos. 52275385 and U2167216).

Recommended articles

References

[1] C. Prakash, S. Singh, C.I. Pruncu, V. Mishra, G. Królczyk, D.Y. Pimenov, A. Pramanik
Surface modification of Ti-6Al-4V alloy by electrical discharge coating process using partially sintered Ti-Nb electrode
Materials, 12 (7) (2019), p. 1006, 10.3390/ma12071006 ↗
View in Scopus ↗ Google Scholar ↗

[2] J.J. Dai, J.Y. Zhu, C.Z. Chen, F. Weng
High temperature oxidation behavior and research status of modifications on improving high temperature oxidation resistance of titanium alloys and titanium aluminides: a review
J. Alloys Compd., 685 (2016), pp. 784-798, 10.1016/j.jallcom.2016.06.212 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[3] H. Koizumi, Y. Takeuchi, H. Imai, T. Kawai, T. Yoneyama
Application of titanium and titanium alloys to fixed dental prostheses
J. Prosthodont. Res., 63 (2019), pp. 266-270, 10.1016/j.jpor.2019.04.011 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[4] X.M. Zha, H. Qin, Z. Yuan, L.Q. Xi, X. Chen, Y. Li, Q.S. Jiang, Z.L. Xu, F. Jiang
Surface integrity and fatigue properties of Ti–6Al–4V alloy under the ultrasonic surface rolling process excited strain rate effect
J. Mater. Res. Technol., 35 (2024), pp. 416-434, 10.1016/j.jmrt.2025.01.024 ↗
Google Scholar ↗

[5] A.V. Yumashev, A.S. Utyuzh, M.V. Mikhailova, V.O. Samusenkov, I.R. Volchkova
Selecting clinical and laboratory methods of manufacture of orthopaedic titanium alloy structures using a biopotentiometer
Curr. Sci. India, 114 (2018), pp. 891-896
https://www.jstor.org/stable/26495251 ↗
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[6] H. Koizumi, Y. Takeuchi, H. Imai, T. Kawai, T. Yoneyama
Application of titanium and titanium alloys to fixed dental prostheses
J. Prosthodont. Res., 63 (2019), pp. 266-270, 10.1016/j.jpor.2019.04.011 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[7] Y. Zhang, Y. Chen, J. Zhou, D. Sun, H. Li
Compound connection mechanism of Al_2O_3 ceramic and TC4 Ti alloy with different joining modes

Sci. Rep-UK, 11 (2021), Article 21251, 10.1038/s41598-021-00815-4 ↗

Google Scholar ↗

[8] J.B. Xia, K. Ren, Y.G. Wang
Reversible joining of zirconia to titanium alloy
Ceram. Int., 45 (2019), pp. 2509-2515, 10.1016/j.ceramint.2018.10.180 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[9] V. Fiore, F. Di Franco, R. Miranda, M. Santamaria, D. Badagliacco, A. Valenza
Effects of anodizing surface treatment on the mechanical strength of aluminum alloy 5083 to fibre reinforced composites adhesive joints
Int. J. Adhesion Adhes., 108 (2021), Article 102868, 10.1016/j.ijadhadh.2021.102868 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[10] X.Q. Cai, D.P. Wang, F.G. Qi, Y. Wang, Q.W. Qiu, Z.W. Yang
Joining of titanium diboride-based ultra high-temperature ceramics to refractory metal tantalum using diffusion bonding technology
J. Eur. Ceram. Soc., 42 (2022), pp. 344-353, 10.1016/j.jeurceramsoc.2021.10.033 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[11] G.B. Niu, D.P. Wang, Z.W. Yang, Y. Wang
Microstructure and mechanical properties of Al2O3/TiAl joints brazed with B powders reinforced Ag-Cu-Ti based composite fillers
Ceram. Int., 43 (2017), pp. 439-450, 10.1016/j.ceramint.2016.09.178 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[12] S.W. Park, H. Lee, B.H. Lee, T.H. Kim, K.I. Kim, S.A. Hong, K.T. Kim
Effect of interface microstructure on joint strength of zirconia/titanium alloy brazed with amorphous Zr-Ti-Ni-Cu active fller metal
Metals, 10 (2020), p. 718, 10.3390/met10060718 ↗

Google Scholar ↗

[13] E. Axinte
Glasses as engineering materials: a review
Mater. Des., 32 (2011), pp. 1717-1732, 10.1016/j.matdes.2010.11.057 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[14] Q. Sun, L.D. Yang, W.C. Yang, H.J. Ji, M.Y. Li, Y.F. Li
Microstructure evolution and bonding mechanism of ZrO_2 ceramic and Ti-6Al-4V alloy joints brazed by Bi_2O_3-B_2O_3-ZnO glass paste
J. Eur. Ceram. Soc., 42 (2022), pp. 5953-5963, 10.1016/j.jeurceramsoc.2022.06.016 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[15] F. He, J.S. Cheng, D.W. Deng, J. Wang
Structure of Bi_2O_3-ZnO-B_2O_3 system low-melting sealing glass
J. Cent. South. Univ., 17 (2010), pp. 257-262
https://link-springer-com.kmclibproxy.khu.ac.kr/article/10.1007/s11771-010-0039-x ↗
Crossref ↗ View in Scopus ↗ Google Scholar ↗

[16] W.J. Qian, P. Chen
Study on the Properties of Bi_2O_3-B_2O_3-BaO Lead-free Glass Using in the Electronic Pastes, 36 (2010), pp. 304-308, 10.1134/S1087659610030053 ↗
View in Scopus ↗ Google Scholar ↗

[17] Z. Wang, Z. Gao, J. Chu, D. Qiu, J. Niu
Low temperature sealing process and properties of kovar alloy to DM305 electronic glass
Metals, 10 (2020), p. 941, 10.3390/met10070941 ↗
View in Scopus ↗ Google Scholar ↗

[18] P.P. Lin, T.S. Lin, P. He, D.P. Sekulic
Wetting behavior and bonding characteristics of bismuth-based glass brazes used to join Li-Ti ferrite systems
Ceram. Int., 43 (2017), pp. 13530-13540, 10.1016/j.ceramint.2017.07.059 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[19] H.Y. Chen, X.W. Ren, G. Wei, Z.K. Shen, X.C. Liu, C. Jian
Effects of glass composition and joining parameters on microstructural evolution and mechanical properties of Al_2O_3/Cu joints brazed with Bi_2O_3-B_2O_3-ZnO glass
J. Manuf. Process., 56 (2020), pp. 735-745, 10.1016/j.jmapro.2020.05.035 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[20] F. Ji, Y.X. Li, W.X. Dong, H. Chen
Effect of the pre-oxidation process on the surface properties of the Ti6Al4V and the wettability of sealing glass
J. Mater. Res. Technol., 30 (2024), pp. 8054-8066, 10.1016/j.jmrt.2024.05.204 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[21] F.X. Zhu, H.W. Gan, H.B. Li, H.S. Hao
Preparation of $Bi_4Ti_3O_{12}$/$CuBi_2O_4$ composite material and their photocatalytic performance
Journal of Dalian Polytechnic University, 43 (2024), pp. 403-409, 10.19670/j.cnki.dlgydxxb.2024.0603 ↗
View in Scopus ↗ Google Scholar ↗

[22] Z.X. Liu, Y.R. An, W.B. Zhang, L.J. Zhu, G.Q. Zhu
Au nanoparticles modified oxygen-vacancies-rich $Bi_4Ti_3O_{12}$ heterojunction for efficient photocatalytic NO removal with high selectivity
J. Alloys Compd., 942 (2023), Article 169018, 10.1016/j.jallcom.2023.169018 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[23] Y.T. Yan, J.H. Lin, K.K. Huang, X.H. Zheng, L. Qiao, S.D. Liu, J. Cao, S.C. Jun, Yusuke Yamauchi, J.L. Qi
Tensile strain-mediated spinel ferrites enable superior oxygen evolution activity
J. Am. Chem. Soc., 145 (44) (2023), pp. 24218-24229, 10.1021/jacs.3c08598 ↗
View in Scopus ↗ Google Scholar ↗

[24] D.Y. Lin, J.X. Hu, R.H. Wu, Y.Z. Liu, X.Q. Li, M.J. SaGong, C.W. Tan, X.G. Song, H.S. Kim
Multiscale plastic deformation in additively manufactured FeCoCrNiMox high-entropy alloys to achieve strength–ductility synergy at elevated temperatures

Int. J. Plast., 183 (2024), Article 104142, 10.1016/j.ijplas.2024.104142 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[25] Y.T. Yan, P.X. Li, Z.L. Zhang, Y.X. Wang, J.K. Zhang, L. Qian, J. Cao, J.L. Qi
Interfacial Si–O coordination for inhibiting the graphite phase enables superior SiC/Nb heterostructure joining by AuNi
Compos. Part B-Eng., 282 (2024), Article 111557, 10.1016/j.compositesb.2024.111557 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[26] M.Q. Liu, L. Hu, X.R. Kang, Y.K. Zhang, X. Liu, L.Y. Xu, Y.D. Han
Deformation mechanism of defect-free Ni50Ti50 alloys via laser powder bed fusion
J. Mater. Sci. Technol., 225 (2025), pp. 111-124, 10.1016/j.jmst.2024.10.051 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

Cited by (2)